

# CONSTRUCTSHUB / BANABUILD

*Construction E-Commerce & Services Marketplace*

## SOFTWARE REQUIREMENTS SPECIFICATION

Web Platform + Cross-Platform Mobile App  
iOS • Android • HarmonyOS

| Field            | Details                                      |
|------------------|----------------------------------------------|
| Document Version | 1.0                                          |
| Status           | Draft – Full Requirements                    |
| Date             | April 2026                                   |
| Domain           | Construction Materials & Services E-Commerce |
| Audience         | Engineering, Product, Design, QA, Legal      |

# 1. Introduction & Project Overview

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## 1.1 Purpose

This Software Requirements Specification (SRS) defines the full functional, non-functional, interface, and business requirements for ConstructsHub / BanaBuild — a specialised construction e-commerce marketplace and companion cross-platform mobile application. The platform connects individual consumers, professional contractors, construction material vendors, skilled technicians, and delivery service providers within a single, integrated digital ecosystem.

## 1.2 Scope

The platform covers the following high-level capabilities:

- Discovery and purchase of construction materials and products (tiles, pipes, paints, cement, steel, fixtures, etc.)
- On-demand matching with skilled technicians and tradespeople (plumbers, painters, electricians, masons, etc.)
- Delivery logistics for both small parcels and heavy bulk material (requiring multi-tonne vehicles)
- Multi-party payment processing with automatic fund disbursement to merchant, technician, and delivery provider
- Post-transaction review, analytics, and financial dashboards for all user types

## 1.3 Target User Types

| User Type           | Role                                             | Key Actions                                                             |
|---------------------|--------------------------------------------------|-------------------------------------------------------------------------|
| Consumer / Buyer    | End customer buying materials or hiring services | Browse, purchase, hire, track, review, pay                              |
| Merchant / Vendor   | Store or supplier selling construction materials | List products, manage orders, review buyers, receive payments           |
| Technician / Worker | Skilled tradesperson offering services           | Register skills, receive quotation requests, submit proposals, get paid |
| Delivery Provider   | Vehicle owner or driver for logistics            | Accept delivery jobs, track deliveries, get paid, review clients        |
| Platform Admin      | Internal operators managing the marketplace      | Moderate listings, resolve disputes, run analytics, manage payouts      |

## 1.4 Definitions & Acronyms

| Term          | Definition                                                            |
|---------------|-----------------------------------------------------------------------|
| SRS           | Software Requirements Specification                                   |
| FR            | Functional Requirement                                                |
| NFR           | Non-Functional Requirement                                            |
| Basket / Cart | Temporary in-session container for selected products                  |
| Checkout      | Multi-step process converting basket items into a paid order          |
| Quotation     | A formal cost estimate submitted by a technician or delivery provider |
| Milestone     | A defined phase of work within a technician's project plan            |
| Disbursement  | Automated transfer of funds to the appropriate payee account          |
| SKU           | Stock-keeping unit – a unique identifier per product variant          |
| ETA           | Estimated Time of Arrival                                             |
| KYC           | Know Your Customer – identity verification process                    |
| API           | Application Programming Interface                                     |

## 2. System Architecture Overview

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### 2.1 Platform Components

- Web Application: Responsive SPA (React / Next.js recommended) accessible via modern browsers
- Mobile Application: Cross-platform (React Native or Flutter) targeting iOS, Android, HarmonyOS
- Backend API: RESTful + GraphQL microservices, containerised (Docker / Kubernetes)
- Database: Relational DB (PostgreSQL) for transactional data + Search Engine (Elasticsearch) for product search
- Maps & Location: Google Maps Platform (Places, Geocoding, Distance Matrix, Directions, Live Tracking)
- Payment Gateway: Primary card processing (Stripe / HyperPay) + in-platform wallet
- Notification Service: Push, email, SMS, and in-app notifications
- File Storage: CDN-backed object storage (AWS S3 / Cloudflare R2) for images, documents
- AI/ML Services: Search relevance ranking, delivery vehicle recommendation engine, fraud detection

### 2.2 High-Level User Journey Map

The core consumer journey follows five major stages:

1. Search & Discovery: Consumer searches for materials using keywords and filters
2. Product Selection: Consumer evaluates product cards, reviews, pricing, and availability
3. Service Attachment: Consumer optionally hires a technician and/or selects a delivery mode
4. Checkout & Payment: Consumer completes purchase with a single multi-party payment
5. Post-Transaction: Tracking, review, and financial settlement across all parties

#### Design Principle

All three commercial parties — merchant, technician, and delivery provider — may be involved in a single transaction. Payment is collected once from the buyer and disbursed separately to each party upon their obligation fulfilment milestone.

## 3. User Accounts & Registration

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### 3.1 Consumer Account

#### 3.1.1 Registration Fields

- Full name, email address, mobile number (OTP-verified)
- Password (hashed, min 8 chars, 1 uppercase, 1 number, 1 symbol)
- Profile photo (optional)
- Nationality and preferred language

#### 3.1.2 Address & Location Management

After registration the consumer may add one or more saved addresses:

- Manual address entry (street, building, city, postal code, country)
- Interactive Google Maps pin-drop: the user positions a draggable marker on the map to set precise coordinates; the system reverse-geocodes to produce a human-readable address label
- Multiple addresses supported (e.g., Home, Office, Site); one marked as default
- Addresses are editable and deletable at any time from the account settings panel

### 3.2 Merchant / Vendor Account

#### 3.2.1 Registration Fields

- Business or trading name, commercial registration number
- Owner full name, email, mobile (OTP-verified)
- Store logo and cover photo
- Physical store address with Google Maps pin-drop confirmation
- Operating hours per day of week
- Bank account details for disbursement (IBAN, account name, bank name)
- KYC documents: trade licence, VAT certificate, ID of authorised signatory

#### 3.2.2 Product & Inventory Capabilities

- Create, edit, and archive product listings (see Section 5)
- Define delivery vehicle specifications per product (see Section 9)
- Manage stock levels and receive low-stock alerts
- Approve or reject incoming orders within a configurable SLA window

## **3.3 Technician / Worker Account**

### **3.3.1 Registration Fields**

- Full name, profile photo, mobile (OTP-verified), email
- Trade categories (multi-select): Plumber, Painter, Electrician, Tiler, Carpenter, Mason, AC Technician, General Handyman, etc.
- Years of experience, certifications (uploadable documents)
- Pricing model: Hourly Rate, Fixed Price per Job, or Quotation Only
- Current availability calendar (weekly schedule with on/off slots)
- Service radius in km from home location (set via Google Maps pin-drop)
- Bank account details for disbursement
- Portfolio images (up to 20 before/after photos)

## **3.4 Delivery Provider Account**

### **3.4.1 Registration Fields**

- Full name, driving licence number, mobile (OTP-verified), email
- Vehicle type (motorcycle, saloon car, pickup truck, light truck, heavy truck, flatbed)
- Vehicle make, model, year, plate number
- Maximum payload capacity (kg) and maximum load volume (m<sup>3</sup>)
- Operating zone (polygon selection on map or radius from base location)
- Bank account details for disbursement

## 4. Product Search & Discovery

### 4.1 Search Bar Behaviour

The global search bar is prominently placed on the homepage header and persists on all inner pages.

- Accepts free-text keyword input (e.g., 'wash sink', 'Italian sink', 'indoor wall paint beige oily', 'grout', 'steel rebar 16mm')
- Autocomplete suggestions appear after 2 characters, drawing from product names, categories, brand names, and popular searches
- Voice search input supported on mobile app
- Search initiated by pressing Enter, tapping the search icon, or selecting an autocomplete suggestion
- Search query is sent to the Elasticsearch index with fuzzy matching, synonym expansion (e.g., 'sink' matches 'basin', 'washbasin'), and category boosting

### 4.2 Search Results Page Layout

Results are displayed in a responsive matrix grid. The default is 4 columns on desktop, 2 on tablet, and 1 on mobile. Each product card contains:

| Card Element              | Details                                                                           |
|---------------------------|-----------------------------------------------------------------------------------|
| Product Image             | Primary image, tappable for full gallery carousel                                 |
| Product Name              | Truncated to 2 lines; full name on hover/tap                                      |
| Brief Description         | Up to 3 lines of key product highlights                                           |
| Price                     | Unit price with currency; 'Get Quotation' badge if price not set                  |
| Vendor / Store Name       | Linked to vendor store page                                                       |
| Review Score              | Star rating (0–5) + number of reviews                                             |
| Availability              | 'In Stock', 'Available in N days', or 'Out of Stock' badge                        |
| Buy Now Button            | Direct one-click purchase (skips basket)                                          |
| Add to Basket Icon        | Adds to current session basket; counter badge updates                             |
| Store Location Icon       | Tapping opens a Google Maps mini-card showing store pin and distance              |
| Delivery / Pickup Toggle  | Quick selector: 'Deliver to me' or 'Pick up from store'                           |
| Technician Available Icon | Green badge if at least one registered technician linked to this product / vendor |
| Get Quotation Button      | Shown when price is hidden; triggers quotation request workflow                   |

## 4.3 Search Filters & Sort Options

A collapsible filter panel appears on the left (desktop) or as a bottom sheet (mobile). Authenticated users have all sort options; guests have basic price and relevance only.

### Sort Options

- Most Relevant (default – AI-ranked by query match, popularity, and proximity)
- Price: Low to High / High to Low
- Nearest Store (uses consumer's default address coordinates)
- Most Popular (by number of sales in last 30 days)
- Highest Review Score
- Technician Available (products where a linked technician is currently available)
- Shortest Delivery Time
- Earliest Availability (for backordered items)

### Filter Options

- Category (multi-select tree: Plumbing, Electrical, Paints, Structural, Flooring, etc.)
- Price range (dual-handle slider)
- Brand / Manufacturer
- Material / Composition
- Colour (swatch picker where applicable)
- Delivery option (Delivery available / Pickup only)
- Technician support available (toggle)
- Store rating (min stars)
- Distance from my location (radius slider in km)
- Availability (In stock now / Within 3 days / Within 7 days)

## 5. Product Detail Page

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### 5.1 Page Layout

When a user taps a product card, a dedicated product detail page opens containing:

#### 5.1.1 Media Gallery

- Full-width image carousel (swipeable); thumbnail strip below
- 360° view or video embed if provided by vendor
- Zoom on hover (desktop) / pinch-to-zoom (mobile)

#### 5.1.2 Product Information Panel

- Full product name, SKU / model number, brand
- Complete description (rich text with support for tables, lists, specification sheets)
- Technical data sheet (PDF download link if provided)
- Price per unit / per kg / per m<sup>2</sup> as applicable; volume discount tiers if configured
- 'Get Quotation' button replaces price when vendor chooses not to display pricing publicly
- Quantity selector (+ / – spinner); minimum order quantity enforced
- Add to Basket and Buy Now buttons

#### 5.1.3 Delivery & Pickup Panel

- 'Deliver to my address' option (shows ETA and estimated delivery cost based on user's saved address)
- 'Pick up from store' option (shows store address, hours, and map mini-card)
- For heavy / bulk items: vehicle requirement calculator (see Section 9.2)

#### 5.1.4 Technician Availability Banner

- If the vendor has linked technicians (or the product category has registered technicians), a banner displays: 'Technician available for installation — from [rate] / hr'
- 'Find a Technician' CTA button navigates to the technician listing page pre-filtered by trade type matching the product category

#### 5.1.5 Vendor Profile Summary

- Store logo, name, city, overall rating, number of reviews
- 'Visit Store' and 'Message Vendor' buttons

#### 5.1.6 Reviews & Ratings Section

- Average star rating with breakdown (1–5 star histogram)
- Individual reviews with reviewer name, date, star rating, comment, and up to 3 photos
- Pagination (10 reviews per page)
- Report review button



## 6. Basket, Checkout & Payment

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### 6.1 Shopping Basket

The basket is a persistent session container accessible via the basket icon in the top navigation bar. A badge counter shows the number of distinct items.

#### 6.1.1 Basket Page Elements

- Line item list: product thumbnail, name, variant, unit price, quantity adjuster (+ / - / type), line total, remove button
- Delivery method per item: 'Deliver' or 'Pick Up' (can be changed here)
- 'Need a Technician?' reminder link beside applicable items
- Price summary panel: subtotal, delivery estimate, technician fees (if hired), taxes, grand total
- Discount / coupon code field
- 'Continue Shopping' link and 'Proceed to Checkout' button

### 6.2 Checkout Flow

Checkout is a linear multi-step wizard:

#### Step 1 – Order Review

- Displays all basket items, quantities, and current prices
- Consumer can remove items or adjust quantities; basket recalculates in real time
- Shows linked technician and delivery selections (editable)
- 'Continue' advances to Step 2

#### Step 2 – Delivery & Address Confirmation

- For items marked 'Deliver': consumer selects from saved addresses or adds a new one; Google Maps embedded picker available
- For items marked 'Pick Up': store address and operating hours confirmed
- For heavy items: vehicle selection (see Section 9.3)
- Consolidated or split delivery options shown where multiple vendors are involved

#### Step 3 – Order Summary & Payment

- Full order summary with line items, service fees, delivery charges, technician deposit (if applicable), taxes
- Grand total clearly displayed broken down by payee: merchant subtotal, technician fee, delivery fee

## 6.3 Payment Options

| Payment Method                           | Trigger Condition                                      | Flow                                                  |
|------------------------------------------|--------------------------------------------------------|-------------------------------------------------------|
| Pay in Platform (Wallet / Bank Transfer) | Any order                                              | Amount reserved; disbursed upon fulfilment milestones |
| Credit / Debit Card                      | User selects card option                               | Card detail entry → 3-D Secure → confirmation page    |
| Cash on Pickup                           | Pickup orders only (optional, merchant-configurable)   | Order confirmed; payment collected at store           |
| Cash on Delivery                         | Delivery orders only (optional, merchant-configurable) | Driver collects cash; driver confirms receipt in app  |
| Additional methods (future)              | Buy Now Pay Later, mobile wallets, etc.                | To be defined in Phase 2                              |

### 6.3.1 Credit Card Entry Page

- Fields: Cardholder name, card number (masked display), expiry month/year, CVV
- Card type auto-detected (Visa, Mastercard, AMEX, Mada, etc.) with logo shown
- 'Save this card for future purchases' toggle (tokenised storage; no raw card data stored)
- 'Pay [Amount]' button; spinner shown during processing
- On 3-D Secure challenge: redirect to bank authentication page, return to platform on completion
- On success: Payment Confirmation Page (see 6.4)
- On failure: inline error message with retry option or 'Choose another payment method'

## 6.4 Payment Confirmation Page

- Order reference number (large, copyable)
- Summary of what was purchased, from which stores
- Expected delivery date(s) or pickup window(s)
- Technician booking confirmation (if applicable)
- 'Track my Order' button (navigates to live tracking, Section 10)
- 'Download Receipt' (PDF)
- Automatic redirect after 5 seconds to the Post-Purchase Review page (Section 11)

## 7. Technician Hiring Module

### 7.1 Technician Discovery Page

Accessed when the consumer taps 'I need a Technician to fix it' from the product page or basket. The page is pre-filtered by the trade type matching the product (e.g., 'Plumber' for sink products, 'Painter' for paint products).

#### 7.1.1 Technician Card Elements

| Card Element              | Details                                                                      |
|---------------------------|------------------------------------------------------------------------------|
| Profile Photo             | Circular avatar; default placeholder if not set                              |
| Full Name & Trade Badge   | Name + coloured trade category chip                                          |
| Completed Jobs Count      | Numeric count of verified completed jobs on the platform                     |
| Pricing Display           | Hourly rate, fixed job price, or 'Get Quotation' button                      |
| Client Satisfaction Score | Star rating + percentage satisfaction + total reviews                        |
| Availability Status       | Green: Available Now   Yellow: Available from [date]   Red: Busy             |
| Location                  | Neighbourhood / city + distance from consumer's default address              |
| ETA to Consumer           | Estimated travel time from technician's current location                     |
| Store Affiliation Badge   | If the technician is registered with the same vendor as the selected product |
| Hire Now Button           | Initiates direct hire (for fixed-price technicians)                          |
| Get Quotation Button      | Triggers quotation request workflow (see 7.2)                                |

#### 7.1.2 Sort & Filter Options for Technicians

- Default sort: nearest location first, then store-affiliated technicians
- Sort by: Cost/Rate (low–high), Review Score, Availability, Distance, ETA
- Filter by: Trade category, Max hourly rate, Availability (today / this week), Minimum review score, Store affiliation

### 7.2 Quotation Workflow

#### 7.2.1 Consumer Sends Quotation Request

- Consumer taps 'Get Quotation'; platform auto-generates a quotation request letter
- Auto-generated letter includes: item name, product specification, purchase quantity, delivery address, and a default task description (e.g., 'Installation and assembly of wash basin' or 'Painting indoor wall with [paint spec]')

- Consumer can edit the auto-generated text and attach photos before sending
- Quotation sent to technician's in-platform inbox and email

### 7.2.2 Technician Receives Quotation Request

- Push notification + email alert to technician
- Technician inbox shows the request with product details, task description, consumer location, and a 'Contact Client' button
- 'Contact Client' opens an in-platform chat thread (see 7.3)

### 7.2.3 Technician Submits Proposal

After reviewing and optionally chatting, the technician creates a formal work proposal:

- Project title and description
- Phase / milestone table: Phase name, description, duration, cost
- Proposed start date and total duration
- Total cost (sum of milestone costs)
- Payment schedule per milestone
- Validity period of the quotation (e.g., valid for 7 days)
- Attach supporting documents or images

### 7.2.4 Consumer Responds to Proposal

- Consumer receives in-platform notification and email with proposal details
- Consumer can: Accept | Request Negotiation (counter-proposal) | Decline
- Negotiation supports multiple counter-proposal rounds, each logged in the chat thread
- On acceptance: technician receives 'Hired' notification; milestone payment schedule is locked
- Platform reserves the total technician fee from consumer's payment; releases per milestone completion

## 7.3 In-Platform Communication

- Persistent chat thread per job between consumer and technician
- Supports text, images, documents (PDF, DOCX), and voice messages
- All messages timestamped and archived for 3 years (dispute resolution)
- Email mirroring: each chat message is also delivered to both parties' email addresses
- Message read receipts shown

## 7.4 Job Execution & Milestone Tracking (Upwork-style)

- Active jobs appear in both consumer and technician dashboards under 'Active Projects'

- Technician marks each milestone as 'Submitted for Review' and uploads evidence (photos, completion report)
- Consumer has 5 business days to approve or raise a dispute on each milestone
- Auto-approval after 5 business days of no response
- On milestone approval: platform releases that milestone's payment from escrow to technician's account
- On final milestone approval: job marked 'Completed'; both parties prompted to leave reviews
- Dispute resolution: admin mediates; decision is final and binding on payment release

## 8. Delivery Module — Light & Standard Goods

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### 8.1 Delivery Option Selection

When the consumer selects 'Deliver to my address' from the product or basket pages, the platform determines the appropriate delivery mode based on total order weight and volume.

#### 8.1.1 Delivery Provider Cards

- Provider name and photo (individual driver) or company logo (fleet)
- Vehicle type icon (motorcycle, car, pickup, truck, heavy truck)
- Current location on mini-map + distance from merchant and consumer
- Estimated delivery time
- Delivery cost
- Provider rating (stars + reviews)
- 'Select This Provider' button

#### 8.1.2 Sort & Filter Options

- Default sort: by ETA ascending
- Sort by: Cost, ETA, Provider Rating, Distance
- Filter by: Vehicle type, Max delivery cost, Max ETA, Minimum provider rating

### 8.2 Live Tracking (Light Goods)

For items classified as light / standard goods (handled by motorcycle or car):

- After dispatch confirmation by merchant, the consumer's tracking page activates
- Google Maps live view showing: merchant pin (origin), consumer pin (destination), and driver animated icon moving in real time (updates every 15 seconds via WebSocket)
- Status timeline: Order Confirmed → Merchant Preparing → Driver Picked Up → On the Way → Delivered
- ETA countdown shown below the map
- Driver contact button (masked phone call through platform relay)
- Consumer can share tracking link with another person

## 9. Delivery Module — Heavy & Bulk Goods

### 9.1 Bulk Goods Classification

Products are classified as 'bulk / heavy goods' when the order meets any of the following criteria:

- Total gross weight exceeds 500 kg
- Total volume exceeds 2 m<sup>3</sup>
- Product is listed by the vendor as a bulk material (cement, steel rebars, gravel, sand, bricks, etc.)

### 9.2 Vehicle Requirement Calculator — Merchant Side (Product Listing)

When a merchant lists a heavy product, they must configure the vehicle specification matrix in the product dashboard:

- Unit weight (kg per piece / bag / reel / ton)
- Unit volume (m<sup>3</sup> per piece)
- Minimum order quantity and maximum order quantity
- For each vehicle type (motorcycle, car, light pickup, medium truck, heavy truck, flatbed trailer): maximum permissible load (kg) and volume (m<sup>3</sup>) and cost per km or per trip

The system uses this data to automatically recommend the minimum suitable vehicle when a consumer places a specific quantity order. Example rules embedded in the algorithm:

#### Example Rule

Consumer orders 2 tins of paint (approx. 4 kg total) → System recommends motorcycle or car. Consumer orders 200 tins (approx. 400 kg) → System recommends light pickup. Consumer orders 30 tonnes of steel rebar → System recommends flatbed trailer + confirms merchant must coordinate with heavy vehicle operator.

### 9.3 Vehicle Selection Page — Consumer Side

When a bulk/heavy delivery is detected, the consumer is redirected to a dedicated vehicle selection page:

- System-recommended vehicle type is highlighted with a 'Best Match' badge
- All eligible vehicle types listed with: provider photo/logo, vehicle specs, payload capacity, ETA, delivery cost, provider rating
- Consumer can override the recommendation (with a warning if selecting an undersized vehicle)
- Filter and sort options mirror those in Section 8.1.2
- 'Confirm Vehicle & Proceed' button advances to the payment page

## 9.4 Heavy Goods Tracking

Tracking for heavy deliveries functions identically to light-goods tracking (Section 8.2) with the following additions:

- Vehicle type icon reflects the actual truck/flatbed (not generic car icon)
- If delivery spans multiple trips (e.g., 40 tonnes of cement across 3 truck loads), each trip is shown as a sub-shipment with its own tracking line
- Merchant and consumer both receive ETA alerts 30 minutes before arrival
- Driver marks arrival on the app; consumer confirms receipt (signature or photo capture)

## 10. User Dashboards

### 10.1 Consumer Dashboard

#### 10.1.1 Overview Panel

- Greeting card with name and profile photo
- Quick stats row: Active Orders, Active Jobs (technician), Saved Items, Loyalty Points
- Recent activity feed (last 5 transactions)

#### 10.1.2 Orders Tab

- List of all orders (sortable, filterable by status and date)
- Order card: order number, product thumbnail(s), merchant name, order date, status, grand total
- 'Track Order' shortcut (for in-progress deliveries)
- 'Reorder' button (adds same items to basket)

#### 10.1.3 Projects Tab (Technician Jobs)

- List of active and completed technician engagements
- Per project: technician name, trade, job title, current milestone, next payment milestone amount
- 'Open Project Chat' and 'View Proposal' links

#### 10.1.4 Payments & Transactions Tab

| Sub-section         | Content                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| Transaction History | Chronological list of all payments: date, order/job ID, payee (merchant / technician / driver), amount, status |
| Payment Methods     | Saved cards (masked), default card toggle, add/remove card                                                     |
| Wallet Balance      | If platform wallet is used: available balance, pending, withdrawal history                                     |
| Receipts            | Downloadable PDF receipts for each transaction                                                                 |
| Analytics Summary   | Monthly spend by category (bar chart), spend trend (line chart), most purchased product categories (pie chart) |

#### 10.1.5 Addresses & Locations

- Saved address list with edit/delete; default address marker
- Interactive map showing all saved pins

#### 10.1.6 Wishlist / Saved Items

- Grid of saved product cards; 'Add to Basket' from wishlist

## 10.2 Merchant / Vendor Dashboard

### 10.2.1 Overview Panel

- Store name, logo, live rating, and last updated timestamp
- Today's stats row: New Orders, Revenue Today, Pending Shipments, Low-Stock Alerts
- Sales chart: Daily revenue for last 30 days (line chart)

### 10.2.2 Products Tab

- Product list with inline stock level editor, active/inactive toggle, edit button
- 'Add New Product' wizard (see Section 5)
- Bulk CSV import / export

### 10.2.3 Orders Tab

- Incoming order list with consumer name, ordered items, order date, delivery/pickup type, total, and status
- Merchant actions per order: Confirm, Reject, Mark Ready, Mark Dispatched
- Order detail modal: full line items, consumer address, delivery provider assigned, any technician linked

### 10.2.4 Reviews & Ratings Tab

- Received product and store reviews with star rating, comment, and reply option
- Consumer trust rating given by merchant (see Section 11.2)

### 10.2.5 Payments & Analytics Tab

| Sub-section          | Content                                                             |
|----------------------|---------------------------------------------------------------------|
| Revenue Summary      | Total earned, pending disbursement, total withdrawn                 |
| Transaction Log      | Line-by-line record of every order payment received                 |
| Disbursement History | Date, order ID, amount disbursed, bank reference number             |
| Top Products         | Best-selling SKUs by quantity and revenue (table + bar chart)       |
| Sales Funnel         | Impressions → Product Views → Add-to-Cart → Purchase (funnel chart) |
| Customer Map         | Heat map of consumer locations (city-level aggregated)              |

### 10.2.6 Buyer Evaluation Panel

After each completed order, the merchant can rate the consumer:

- Overall experience (1–5 stars)
- Punctuality for pickup (if applicable)
- Communication quality
- Would you sell to this buyer again? (Yes / No / Maybe)
- Optional free-text comment (private, not shown to consumer)

## 10.3 Technician Dashboard

### 10.3.1 Overview Panel

- Profile completion progress bar
- Quick stats: Active Jobs, Pending Quotations, Completed Jobs (lifetime), Total Earned (this month)
- Availability quick-toggle (Available Now / Unavailable)

### 10.3.2 Quotations Tab

- Incoming quotation requests with consumer name, product, task, location, and date received
- Actions: View Details, Contact Client, Submit Proposal, Decline

### 10.3.3 Active Jobs Tab

- Project cards with phase/milestone progress bar
- Actions per milestone: Submit for Review (with photo/document upload), View Consumer Feedback

### 10.3.4 Payments & Analytics Tab

| Sub-section            | Content                                                              |
|------------------------|----------------------------------------------------------------------|
| Earnings Summary       | Total earned, in-escrow (pending milestone release), total withdrawn |
| Transaction Log        | Per milestone payment: job ID, milestone name, amount, date released |
| Monthly Earnings Chart | Bar chart of monthly income for last 12 months                       |
| Job Completion Rate    | Completed vs. cancelled jobs (donut chart)                           |
| Review Score Trend     | Average rating per month (line chart)                                |

## 10.4 Delivery Provider Dashboard

### 10.4.1 Overview Panel

- Vehicle photo, capacity stats, live availability toggle
- Today: Jobs Offered, Jobs Accepted, Jobs Completed, Earnings Today

### 10.4.2 Jobs Tab

- Incoming delivery offers: merchant address, consumer address, cargo description, estimated weight, recommended for this vehicle (Yes/No badge), delivery fee offered
- Actions: Accept, Decline, Counter-offer (for negotiable delivery fees)
- Active delivery: shows live map, pickup confirmation button, delivery confirmation button (with photo capture)

### 10.4.3 Payments & Analytics Tab

| Sub-section      | Content                                                  |
|------------------|----------------------------------------------------------|
| Earnings Summary | Total earned, pending, withdrawn                         |
| Trip Log         | Date, order ID, merchant, consumer, cargo, distance, fee |

| Sub-section           | Content                             |
|-----------------------|-------------------------------------|
| Weekly Earnings Chart | Bar chart (last 8 weeks)            |
| Zone Heat Map         | Areas of highest delivery frequency |

#### 10.4.4 Driver's Consumer Review Panel

After job completion (or auto-close after 5× the estimated delivery time):

- Rate the consumer: on-time availability for pickup receipt, accurate address, respectful interaction
- Flag problematic orders for admin review

## 11. Post-Transaction Review System

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### 11.1 Consumer Review Page (After Purchase)

Automatically shown after payment confirmation. The page is divided into review cards:

#### 11.1.1 Product Review Card

- Product image, name, and vendor
- Star rating (1–5) for product quality
- Star rating for value for money
- Free text review (min 20 characters to submit)
- Photo upload (up to 5 images)
- Would you recommend this product? (Yes / No)

#### 11.1.2 Store / Vendor Review Card

- Store name and logo
- Star ratings for: product accuracy, packaging, communication, overall satisfaction
- Free text review

#### 11.1.3 Delivery Review Card (if applicable)

- Driver name and vehicle type
- Star ratings for: timeliness, cargo condition on arrival, driver professionalism
- Free text comment

#### 11.1.4 Technician Review Card (if job completed)

- Technician name, photo, trade
- Star ratings for: work quality, punctuality, communication, adherence to quoted cost
- Free text review and optional before/after photos

### 11.2 Merchant / Vendor Rates the Consumer

Shown in the merchant dashboard after the order is completed or picked up:

- Overall buying experience rating (1–5 stars)
- Sub-ratings: payment punctuality, communication, pickup punctuality
- 'Would you transact with this buyer again?' toggle
- Private notes (internal, not shown to consumer)
- Consumer trust badge updated on platform based on aggregated merchant feedback

## 12. Payment Processing & Financial Disbursement

### 12.1 Payment Collection

The platform collects a single consolidated payment from the consumer at checkout, regardless of how many parties are involved (merchant, technician, delivery provider). The payment is split internally:

| Payment Component        | Recipient         | Release Trigger                                                          |
|--------------------------|-------------------|--------------------------------------------------------------------------|
| Product amount           | Merchant          | Consumer confirms receipt of goods (or auto-release after 48 hrs)        |
| Delivery fee             | Delivery Provider | Delivery confirmed (photo + consumer acknowledgement)                    |
| Technician milestone fee | Technician        | Consumer approves each milestone (or auto-release after 5 business days) |
| Platform service fee     | Platform          | Deducted from each of the above at point of disbursement                 |

### 12.2 Disbursement Mechanism

- Disbursements are processed via bank transfer (IBAN / SWIFT) or payment gateway payout API
- Each disbursement generates a unique reference number and is logged in the respective dashboard
- Disbursement schedule: within 24 hours of release trigger, subject to banking hours
- Failed disbursements (invalid IBAN, frozen account): admin alerted; funds held pending resolution
- Platform service fee percentage is configurable per merchant tier and per service category

### 12.3 Financial Records & Audit Trail

- Every payment event (collection, hold, release, disbursement, refund) is recorded with: timestamp, amount, currency, parties involved, order/job reference, and processing bank reference
- Immutable audit log accessible to platform admins
- Monthly financial statements auto-generated for each commercial user (merchant, technician, delivery provider) — downloadable as PDF and Excel

- Refund workflow: consumer raises a refund request → merchant approves / rejects within 48 hrs → platform admin arbitrates if unresolved → refund to original payment method within 5–7 business days

## 13. Non-Functional Requirements

| Category      | Requirement                 | Target                                                     |
|---------------|-----------------------------|------------------------------------------------------------|
| Performance   | Page load time (web)        | < 2.5 s on 4G connection (LCP)                             |
| Performance   | Search results return time  | < 800 ms for up to 10,000 concurrent queries               |
| Performance   | Payment processing response | < 3 s end-to-end                                           |
| Scalability   | Concurrent users            | >= 50,000 simultaneous sessions                            |
| Availability  | Platform uptime             | >= 99.9% monthly SLA                                       |
| Security      | Data at rest                | AES-256 encryption                                         |
| Security      | Data in transit             | TLS 1.3 minimum                                            |
| Security      | PCI-DSS compliance          | Level 1 (card data never stored on platform servers)       |
| Privacy       | GDPR / local data laws      | Full compliance; right to erasure supported                |
| Localisation  | Languages                   | Arabic (RTL), English; extensible to French, Urdu          |
| Localisation  | Currencies                  | SAR primary; USD, AED, KWD via settings                    |
| Accessibility | Web (WCAG)                  | WCAG 2.1 AA minimum                                        |
| Mobile        | Supported OS versions       | iOS 15+, Android 10+, HarmonyOS 3+                         |
| Offline       | Mobile app offline mode     | Browse cached products; full features require connectivity |
| Backup        | Database backup frequency   | Daily full + hourly incremental; 30-day retention          |
| Audit         | Log retention               | Transaction logs: 7 years; chat logs: 3 years              |

## 14. Notification System

| Event                          | Recipient(s)                   | Channel(s)                |
|--------------------------------|--------------------------------|---------------------------|
| New order placed               | Merchant                       | Push, email, in-app       |
| Order confirmed by merchant    | Consumer                       | Push, email, SMS          |
| Order dispatched               | Consumer                       | Push, SMS                 |
| Delivery ETA 30 min            | Consumer                       | Push, SMS                 |
| Order delivered / picked up    | Consumer + Merchant            | Push, email               |
| Quotation request received     | Technician                     | Push, email, in-app inbox |
| Proposal submitted             | Consumer                       | Push, email, in-app inbox |
| Proposal accepted / rejected   | Technician                     | Push, email               |
| Milestone submitted for review | Consumer                       | Push, email               |
| Milestone approved / disputed  | Technician                     | Push, email               |
| Job completed                  | Both parties                   | Push, email               |
| Payment disbursed              | Merchant / Technician / Driver | Push, email               |
| Review received                | Any rated party                | Push, email               |
| Low stock alert                | Merchant                       | Push, email               |
| Delivery job offered           | Driver                         | Push, in-app              |
| Message received (chat)        | Relevant party                 | Push, email digest        |

## 15. Platform Administration Module

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### 15.1 Admin Dashboard Overview

- Real-time KPI widgets: GMV (Gross Merchandise Value) today, active orders, new user registrations today, platform revenue today
- Alert panel: pending KYC verifications, open disputes, flagged reviews, failed disbursements

### 15.2 User Management

- Search and view any user account (consumer, merchant, technician, driver)
- Approve or reject KYC submissions with notes
- Suspend, ban, or reactivate accounts
- Impersonate (read-only) for support purposes

### 15.3 Content Moderation

- Review and approve / reject new product listings
- Handle reported reviews (hide, delete, warn reviewer)
- Manage product categories and attribute schemas

### 15.4 Dispute Resolution

- View full case file: order details, payment status, chat logs, milestone evidence
- Render binding decision: release funds to buyer or seller; partial refunds supported
- Log decision rationale (stored for 7 years)

### 15.5 Financial Administration

- View all pending disbursements and approve manual overrides
- Configure platform service fee rates per category and merchant tier
- Export complete transaction ledger (CSV / XLSX)
- Monitor payment gateway health and reconciliation status

### 15.6 Analytics & Reporting

- GMV trend (daily, weekly, monthly, yearly) — line chart
- Category breakdown by revenue — horizontal bar chart
- User growth funnel — funnel chart
- Geographic sales heat map — Google Maps overlay
- Top merchants, technicians, and drivers by revenue and rating tables
- Automated weekly and monthly PDF reports emailed to designated admin recipients



## 16. Future Phase Considerations

| Feature                                               | Rationale                                                                                                               | Suggested Phase |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------|
| Buy Now Pay Later (BNPL)                              | Increase purchasing power for large material orders                                                                     | Phase 2         |
| Mobile wallet integrations (STC Pay, Apple Pay, etc.) | Reduce payment friction on mobile                                                                                       | Phase 2         |
| AI-powered project estimator                          | Consumer describes a project (e.g., 'renovate bathroom 8m <sup>2</sup> '); AI generates material list and cost estimate | Phase 2         |
| AR product preview                                    | Place tiles, paint colours, or fixtures in camera view using augmented reality                                          | Phase 3         |
| B2B contractor accounts                               | Bulk purchasing with invoice-based credit terms for construction firms                                                  | Phase 3         |
| IoT inventory integration                             | Real-time stock level sync from merchant's ERP / warehouse management system                                            | Phase 3         |
| Multi-language AI chat support                        | LLM-powered customer support bot in Arabic and English                                                                  | Phase 2         |
| Subscription loyalty programme                        | Points, cashback, and VIP tiers for frequent buyers                                                                     | Phase 2         |
| Carbon footprint tracking                             | Show estimated CO <sub>2</sub> per delivery; offer offset options                                                       | Phase 4         |

## 17. Appendix — Full Glossary

| Term         | Definition                                                                                                        |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Basket       | Client-side and session-persistent container holding items selected for purchase, prior to checkout               |
| Checkout     | The multi-step purchase completion process: review → address → payment → confirmation                             |
| Disbursement | Automated transfer of collected funds from the platform escrow to the appropriate payee bank account              |
| Escrow       | Temporary holding of consumer-paid funds by the platform until fulfilment conditions are met                      |
| Milestone    | A defined deliverable phase within a technician's project plan, each with its own cost and timeline               |
| Quotation    | A formal cost and scope document submitted by a technician or delivery provider in response to a consumer request |
| SKU          | Stock-Keeping Unit — a unique alphanumeric identifier assigned to each distinct product variant                   |
| GMV          | Gross Merchandise Value — total transactional value of all orders placed on the platform                          |
| KYC          | Know Your Customer — regulatory identity verification for commercial account holders                              |
| OTP          | One-Time Password — a time-limited code sent via SMS or email for identity verification                           |
| LCP          | Largest Contentful Paint — a Core Web Vitals metric for page load performance                                     |
| WCAG         | Web Content Accessibility Guidelines published by the W3C                                                         |
| IBAN         | International Bank Account Number — standardised format for bank account identification                           |
| PCI-DSS      | Payment Card Industry Data Security Standard                                                                      |
| ETA          | Estimated Time of Arrival                                                                                         |
| SLA          | Service Level Agreement — a committed standard of service (e.g., uptime guarantee)                                |
| RTL          | Right-to-Left — text directionality used for Arabic language interfaces                                           |
| CDN          | Content Delivery Network — geographically distributed cache for static assets                                     |
| API          | Application Programming Interface — a defined contract for system-to-system communication                         |
| SPA          | Single Page Application — a web app that loads once and updates dynamically without full page reloads             |